Project Design Phase
Proposed Solution Template

Date	8 September 2026
Team ID	
Project Name	Car Rental Request Automation in ServiceNow
Maximum Marks	2 Marks

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	
2.	Idea / Solution description	
3.	Novelty / Uniqueness	
4.	Social Impact / Customer Satisfaction	
5.	Business Model (Revenue Model)	
6.	Scalability of the Solution	

Proposed Solution – Car Rental Request Automation

Implement a ServiceNow Service Catalog solution that standardizes the complete car rental request lifecycle. Employees submit a single catalog item with required travel and vehicle details. ServiceNow creates a Request/RITM, performs configured validation, obtains approval, routes approved requests to the Transport team, supports availability and fulfillment updates, and notifies the requester at important stages.

Proposed Solution Components

Component	Implementation	Benefit
Service Catalog	Car Rental Request catalog item with clear mandatory variables.	Standardized request submission.
Catalog Variables	Pickup/drop-off locations, travel dates/times, car type, travel purpose, passenger count and remarks.	Complete information for fulfillment.
Request & RITM	Create and track the ServiceNow Request and Requested Item.	Centralized traceability.
Automation	Flow Designer/Workflow controls	Reduces manual coordination.

	validation, approval, assignment, notifications and completion.	
Approval	Route request to configured approver; record approve/reject decision.	Ensures authorization.
Transport Assignment	Automatically route approved RITM to Transport assignment group.	Faster fulfillment handoff.
Availability & Fulfillment	Transport reviews availability and records vehicle/rental details.	Confirms the requested arrangement.
Notifications	Notify requester on submission, approval/rejection and completion.	Improves transparency.

Expected Outcome

The solution provides a transparent, trackable and repeatable car rental process with less manual follow-up, fewer incomplete requests and clear ownership from submission through fulfillment.

Project Design Phase
Solution Architecture

Date	8 September 2026
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Project Name	Car Rental Request Automation in ServiceNow
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Example - Solution Architecture Diagram:

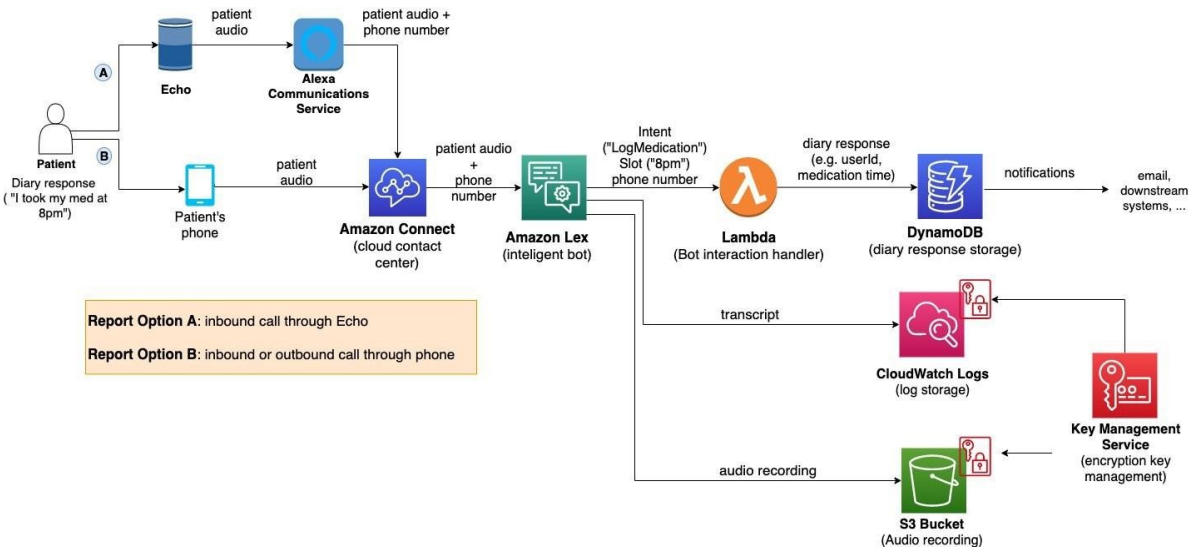


Figure 1: Architecture and data flow of the voice patient diary sample application

Reference: https://aws.amazon.com/blogs/industries/voice-applications-in-clinical-research-powered-by-ai-on-aws-part-1-architecture-and-design-considerations/

Solution Architecture – Car Rental Request Automation

The architecture is centered on ServiceNow. The Service Catalog provides the employee entry point, while Request/RITM records, Flow Designer/Workflow, approvals, assignment groups and notifications coordinate the fulfillment lifecycle.

Logical Architecture

Employee / Requester
↓ ServiceNow Service Catalog – Car Rental Request
↓ Request + RITM Creation / Validation
↓ Flow Designer / Workflow
↓ Approval → Transport Assignment
↓ Availability Check → Fulfillment → Notifications
↓ Request Completed / Tracked

Architecture Components

Layer	ServiceNow Component	Responsibility	Output
Presentation	Service Catalog / Catalog Item	Collect employee request details.	Submitted catalog request
Request Data	Request + RITM + Variables	Store request and variable values.	Trackable RITM
Automation	Flow Designer / Workflow	Control sequence and conditions.	Automated process
Governance	Approval	Obtain authorization.	Approval decision
Fulfillment	Assignment Group / Transport Team	Review availability and arrange vehicle.	Fulfillment details
Communication	Notifications	Inform requester and teams.	Status updates
Security	Roles / ACLs	Restrict access to appropriate users.	Controlled access

Design Principles

- Keep the catalog form simple and focused.
- Capture required information at the point of submission.
- Automate repetitive routing and communication.
- Keep approval and fulfillment traceable through the RITM.
- Apply role-based access and maintain an audit trail.

Project Design Phase
Problem – Solution Fit Template

Date	8 September 2026
Team ID	
Project Name	Car Rental Request Automation in ServiceNow
Maximum Marks	2 Marks

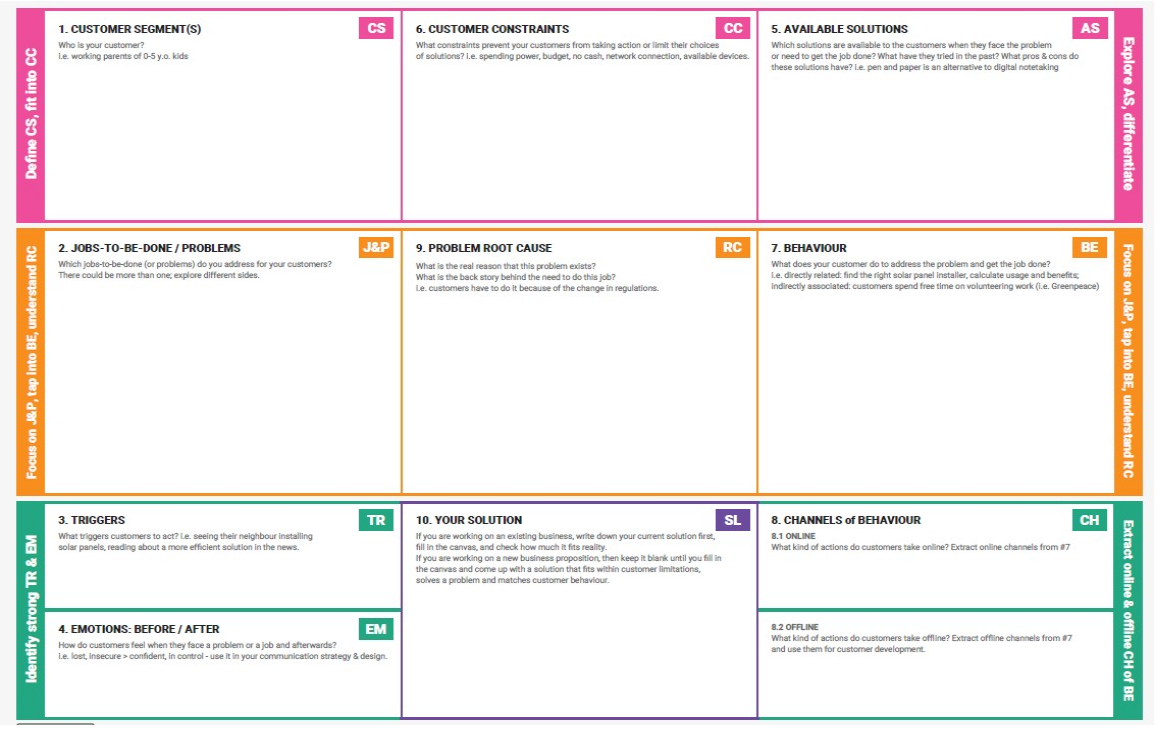
Problem – Solution Fit Template:

The Problem-Solution Fit simply means that you have found a problem with your customer and that the solution you have realized for it actually solves the customer's problem. It helps entrepreneurs, marketers and corporate innovators identify behavioral patterns and recognize what would work and why

Purpose:

- ❑ Solve complex problems in a way that fits the state of your customers.
- ❑ Succeed faster and increase your solution adoption by tapping into existing mediums and channels of behavior.
- ❑ Sharpen your communication and marketing strategy with the right triggers and messaging.
- ❑ Increase touch-points with your company by finding the right problem-behavior fit and building trust by solving frequent annoyances, or urgent or costly problems.
- ❑ **Understand the existing situation in order to improve it for your target group.**

Template:



References:

1. https://www.ideahackers.network/problem-solution-fit-canvas/
2. https://medium.com/@epicantus/problem-solution-fit-canvas-aa3dd59cb4fe

Problem–Solution Fit – Car Rental Request Automation

The proposed ServiceNow solution directly addresses the identified problems of manual car rental requests, incomplete information, unclear approval status, manual assignment and repeated follow-ups.

Problem–Solution Mapping

Identified Problem	Root Cause	Proposed ServiceNow Solution	Expected Improvement
Employees do not have one standard request channel.	Requests may be handled through emails/calls or different formats.	Provide one Car Rental Request catalog item.	Consistent and easy request submission.
Requests may be incomplete.	Required information is not consistently captured.	Use mandatory catalog variables and validation.	Fewer incomplete requests.
Approval status is unclear.	Manual approval communication and follow-up.	Automated approval with status updates.	Better visibility and faster decisions.
Requests must be manually forwarded to Transport.	No automated routing path.	Assign approved RITMs to the Transport assignment group.	Clear ownership and reduced delay.
Vehicle availability is not visible.	Availability is coordinated manually.	Transport records availability and fulfillment details on the RITM.	Traceable fulfillment.
Employees repeatedly ask for updates.	No central status tracking.	RITM status tracking and notifications.	Reduced follow-up and improved user experience.
Management has limited process visibility.	Request data is spread across communication channels.	Centralize requests in ServiceNow for reporting.	Better monitoring and continuous improvement.

Fit Assessment

The solution has strong problem–solution fit because each major user pain point is addressed by a specific ServiceNow capability. The design can also be extended later with additional approval rules, vehicle availability data, reporting dashboards or integrations without changing the basic request lifecycle.